Lecture 4

Covering spaces and rotations

Objectives. *The circle projection of Lecture 3 has local inverse branches. A covering map has the same property over every sufficiently small open set. We lift paths using these branches and recover loop classes from their endpoints above a fixed point. The antipodal covering then computes the fundamental group of the rotation space: a full turn represents the nontrivial class, and two full turns represent the identity.*

By the end of this lecture you will be able to:

- Prove path and path-homotopy lifting for a covering map.
- Recover loop classes from a simply connected covering space.
- Compute $\pi_1(\mathbb{R}\mathrm{P}^n)$ for $n \geq 2$ and $\pi_1(\mathrm{SO}(3))$.
- Distinguish the lifted paths of a 2π and a 4π rotation.

4.1 Coverings and local inverse branches

Definition 4.1 (Covering map). A *covering map* is a continuous surjection $p\colon \widetilde{X} \to X$ such that each $x \in X$ has an open neighbourhood U with

$$p^{-1}(U) = \coprod_{\alpha} V_\alpha, \qquad p|_{V_\alpha} : V_\alpha \longrightarrow U \text{ a homeomorphism.}$$

The symbol $\coprod$ denotes a disjoint union. The sets V_α are open in $\widetilde{X}$. We call U *evenly covered* and each V_α a *sheet over* U.

The fibre $p^{-1}(\{x\})$ is the inverse image of the point, as in Lecture 2. It contains one point in each sheet over U. In particular, the fibre has the discrete subspace topology: every one of its points is open in the fibre. Its cardinality is constant as x varies in an evenly covered set. The figure shows the local structure.

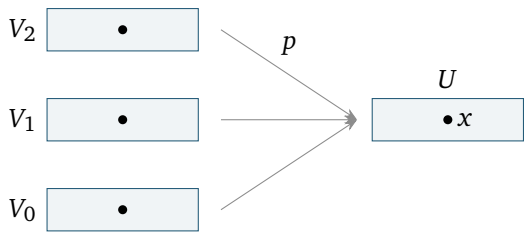


Figure 4.1: Each sheet maps homeomorphically onto the same open set U. The marked points belong to one fibre.

For example, the projection $u \mapsto e^{2\pi \mathrm{i} u}$ from $\mathbb{R}$ to S^1 is a covering. The semicircles U_r

of § 3.3 have inverse images $\coprod_{k\in\mathbb{Z}}(J_r + k)$. Each restriction has inverse $b_r + k$. The map $z \mapsto z^m$ on S^1, for an integer $m \geq 1$, is an m-sheeted covering. An arc of angle less than 2π has m disjoint inverse arcs, obtained by dividing its angles by m and adding $2\pi k/m$ for $k = 0, \ldots, m-1$.

A *local homeomorphism* restricts near each domain point to a homeomorphism onto an open subset of the target. Every covering has this property. The converse fails, even for a surjection. Consider

$$q\colon (0, 3\pi) \longrightarrow S^1, \qquad q(\theta) = e^{\mathrm{i}\theta}.$$

Small intervals give local inverses. The fibre over 1 is $\{2\pi\}$, but the fibre over $e^{\mathrm{i}\varepsilon}$ has two points whenever $0 < \varepsilon < \pi$. Every neighbourhood of 1 contains such points. Thus 1 has no evenly covered neighbourhood.



Figure 4.2: The truncated angle interval contains one preimage of 1, but two of $e^{\mathrm{i}\varepsilon}$. Open endpoints prevent a second branch at 1.

4.2 Lifting paths and their homotopies

Recall that a *lift* of $f\colon Y \to X$ is a continuous map $\tilde{f}\colon Y \to \widetilde{X}$ with $p \circ \tilde{f} = f$. Write $I = [0, 1]$.

> **Theorem 4.2** (Path and path-homotopy lifting). *For a covering p, a path $\gamma\colon I \to X$ has a unique lift after specifying $\tilde{\gamma}(0)$ above $\gamma(0)$. A homotopy $h\colon I^2 \to X$ has a unique lift after a lift of its bottom edge has been specified. If h fixes its path endpoints, so does its lift.*

Proof. First, uniqueness of a path lift follows locally. On a subinterval mapping into an evenly covered U, any lift stays in one sheet: the interval is connected and the sheets are disjoint and open. The sheet is determined by one lifted value. Its inverse branch determines the whole subinterval.

For existence, pull back evenly covered sets to an open cover of I. The Lebesgue-number lemma of Lemma 3.5 gives a finite subdivision such that each closed subinterval maps into one such set. On the first subinterval use the sheet containing the prescribed initial point. On each successive subinterval use the sheet containing the preceding endpoint. The formulas agree at the joins and paste continuously. Local uniqueness on successive subintervals gives uniqueness on I.

For a homotopy, pull the same open cover back to the compact square. Choose a grid with each cell mapping into an evenly covered set. Extend the bottom lift row by row, from left to right. Choose on each cell the inverse branch whose sheet contains the already assigned lower-left value. On its bottom and left edges, path-lift uniqueness forces agreement with every previously assigned value. There are no other previously assigned points on the cell. Finite pasting gives a continuous lift on the square. The same cell order proves uniqueness. This is the grid construction of Lemma 3.6, with sheets in place of integer translates.

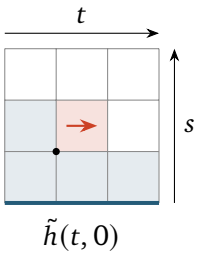


Figure 4.3: The new inverse branch is chosen at the marked corner. Uniqueness on shared edges makes it agree with the shaded region.

Finally, if an edge of h is constant, its lift is a path in a discrete fibre. A continuous image of the connected interval is connected, so this path is constant. Apply this to the two endpoint edges. ∎

In particular, endpoint-fixed homotopic paths have lifts with the same endpoint when their initial lifts agree. The induced homomorphism p_* is injective: lift a based contraction of a projected loop. The lift is a based contraction of the original loop. Also,

$$[\gamma] \in p_*\pi_1(\widetilde{X}, \tilde{x}_0) \quad \Longleftrightarrow \quad \text{the lift of } \gamma \text{ from } \tilde{x}_0 \text{ closes.}$$

For the reverse implication project the closed lift. For the forward implication compare γ with the projection of a representative loop upstairs and lift their based homotopy.

A space is *locally path-connected* if every neighbourhood of each point contains a path-connected open neighbourhood of that point. The following *lifting criterion* is stated with its proof sketched. For path-connected, locally path-connected Y, a continuous map $f\colon (Y, y_0) \to (X, x_0)$ lifts with value $\tilde{x}_0$ at y_0 exactly when

$$f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \tilde{x}_0).$$

Necessity follows from $f_* = p_*\tilde{f}_*$. For sufficiency, choose a path from y_0 to y and lift its image under f. Define $\tilde{f}(y)$ as the lifted endpoint. The inclusion and the closed-lift test make this independent of the chosen path. On a small path-connected open set mapping into an evenly covered set, this rule is one inverse branch composed with f, so is continuous. This is Hatcher, Proposition 1.33.

For example, let $p(z) = z^3$ and $f(z) = z^6$ on S^1, with base point 1. The induced maps multiply winding by 3 and 6 respectively. The inclusion $6\mathbb{Z} \subseteq 3\mathbb{Z}$ permits a lift. It is $\tilde{f}(z) = z^2$, since $(z^2)^3 = z^6$ and $\tilde{f}(1) = 1$. In contrast, $z \mapsto z^5$ has no such lift because $5\mathbb{Z}$ is not contained in $3\mathbb{Z}$. Dividing the angle by three along the generating loop ends at $e^{2\pi i 5/3} \neq 1$ and therefore does not define a loop upstairs. The obstruction concerns a global lift, although every small arc has inverse branches.

4.3 Fibres, loop classes and deck transformations

Definition 4.3 (Deck transformation). A *deck transformation* of p is a homeomorphism $D\colon \widetilde{X} \to \widetilde{X}$ with $p \circ D = p$. Their group under composition is written $\mathrm{Deck}(p)$.

On a path-connected covering space, a deck transformation is determined by its value at one point. Indeed, for a path a from that point to any other point, $D \circ a$ is the unique lift of $p \circ a$ from the specified image point. In particular, a deck transformation fixing one point is

the identity. The deck group acts *transitively* on a fibre if for any two points of that fibre some deck transformation sends the first to the second.

Corollary 4.4 (Counting a simply connected covering). *Let $p\colon \widetilde{X} \to X$ be a covering with $\widetilde{X}$ simply connected. Fix $\tilde{x}_0$ above x_0. Sending a loop class to its lifted endpoint gives a bijection*

$$\pi_1(X, x_0) \longrightarrow p^{-1}(\{x_0\}).$$

If the deck group acts transitively on this fibre, it is isomorphic to $\pi_1(X, x_0)$.

Proof. The endpoint depends only on the based class by homotopy lifting. Every fibre point can be joined to $\tilde{x}_0$, since simple connectedness includes path-connectedness. Projecting this path proves surjectivity. If two lifted paths have the same endpoint, join the first to the reverse of the second. This loop contracts in $\widetilde{X}$. Its projection shows that the two original loop classes agree. This proves injectivity.

Under transitivity, let D_γ be the unique deck transformation taking $\tilde{x}_0$ to the endpoint of the lift $\tilde{\gamma}$. The preceding bijection and deck uniqueness give a bijection of groups. The lift of $\gamma \cdot \delta$ first follows $\tilde{\gamma}$, then $D_\gamma \circ \tilde{\delta}$. It therefore ends at

$$D_\gamma(D_\delta(\tilde{x}_0)), \qquad D_{\gamma\cdot\delta} = D_\gamma \circ D_\delta.$$

Thus the bijection preserves multiplication with the course's first-path-first convention. The isomorphism uses the chosen lifted base point. ∎

For example, the deck transformations of $\mathbb{R} \to S^1$ are exactly $u \mapsto u + k$ for $k \in \mathbb{Z}$. These are deck transformations, act transitively on the fibre, and exhaust the possibilities by uniqueness from the image of 0. They recover $\pi_1(S^1, 1) \cong \mathbb{Z}$.

4.4 The antipodal cover and rotations

Proposition 4.5 (Spheres of dimension at least two · *sketched*). *The sphere S^n is simply connected for $n \geq 2$.*

Proof sketch. Short spherical arcs join nonantipodal points. A path via a third point joins antipodes, so the sphere is path-connected.

First, cover the loop by open hemispheres. Compactness and Lemma 3.5 give a finite subdivision whose pieces each lie in one hemisphere. Replace each piece by the shortest spherical arc joining its endpoints in that hemisphere. Normalizing their straight interpolation gives the homotopy: its scalar product with the hemisphere centre stays positive.

Next, the resulting polygon lies in finitely many two-dimensional linear subspaces. For $n \geq 2$ these are proper subspaces of $\mathbb{R}^{n+1}$ and their union misses a point of the sphere. Finally, stereographic projection from that point identifies its complement with $\mathbb{R}^n$, where the polygon contracts with its base point fixed. The formulas of Lecture 1 apply in n coordinates. This sketches the sphere result of Hatcher, Proposition 1.14, using polygonal approximation. The original continuous loop need not miss a point.

Recall from Lecture 2 that $\mathbb{R}\mathrm{P}^n$ identifies antipodal points of S^n. The quotient map is a covering. To see this, for $x \in S^n$ take the cap $V = \{u \in S^n : u \cdot x > 1/2\}$. The inverse image of its quotient image is $V \coprod (-V)$. The two caps are disjoint. The quotient map is open, because the saturation of any open set W is the open set $W \cup (-W)$. Each cap therefore maps homeomorphically onto the same open quotient set. These sets cover $\mathbb{R}\mathrm{P}^n$.

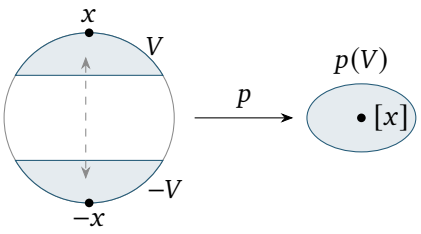


Figure 4.4: The two disjoint caps V and $-V$ lie above the same quotient neighbourhood. The antipodal map exchanges the two sheets.

Write $\mathbb{Z}_2$ for the integers modulo two.

Theorem 4.6 (Projective space and rotations · *proved given S^n simply connected*)**.** *For $n \geq 2$, the antipodal cover gives $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$. Consequently $\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}_2$.*

Proof. The sphere is simply connected by Proposition 4.5, at the sketched level just given. The identity and antipodal map act transitively on the two-point fibre. Deck uniqueness excludes any other transformation. The antipodal map has square the identity, so Corollary 4.4 gives $\mathbb{Z}_2$. The homeomorphism $\mathbb{R}\mathrm{P}^3 \cong \mathrm{SO}(3)$ was proved in Proposition 2.9 by the axis–angle construction. ∎

The circle case is different: $\mathrm{SO}(2) \cong S^1$ has fundamental group $\mathbb{Z}$. The general result $\pi_1(\mathrm{SO}(m)) \cong \mathbb{Z}_2$ for $m \geq 3$ is cited from Nakahara, Table 4.1; only $m = 3$ is proved in this lecture.

4.5 Universal covers and their classification

A simply connected covering space is called a *universal cover*. A space is *semilocally simply connected* if each $x \in X$ has an open neighbourhood U such that every loop in U based at x contracts in X. The neighbourhood itself need not be simply connected. A *covering isomorphism* is a homeomorphism between the total spaces commuting with their projections to X. In the based version it also preserves the chosen lifted point. Two subgroups are *conjugate* if they have the form H and $k^{-1}Hk$ for some group element k.

Theorem 4.7 (Existence and classification · *cited*)**.** *Let X be nonempty, connected, locally path-connected and semilocally simply connected. It has a universal cover. Isomorphism classes of based connected coverings correspond to subgroups of $\pi_1(X, x_0)$ by $p_*\pi_1(\widetilde{X}, \tilde{x}_0)$. Without chosen base points, they correspond to conjugacy classes of subgroups.*

This is Hatcher, Theorem 1.38. Local path-connectedness makes path components open, so connectedness implies path-connectedness here, both downstairs and in a connected cover.

4.6 A full turn and a double turn

We compute the class of rotation through angle $2\pi t$ about the third axis, for $t \in I$. Write a point of S^3 as (a, v), where $a \in \mathbb{R}$, $v \in \mathbb{R}^3$ and $a^2 + |v|^2 = 1$. The corresponding rotation is

$$w \longmapsto (a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w).$$

When $v = 0$, this is the identity. Otherwise, substituting $a = \cos\theta$ and $v = \sin\theta\, n$ gives Lecture 2's rotation $R(2\theta n)$ about the unit axis n. The formula is unchanged by $(a, v) \mapsto (-a, -v)$, so it induces a continuous map from $\mathbb{R}\mathrm{P}^3$ to $\mathrm{SO}(3)$.

To identify that map, choose representatives with $a \geq 0$ and $0 \leq \theta \leq \pi/2$. The corresponding axis–angle vector is

$$v \longmapsto \frac{2 \arcsin |v|}{|v|} v \quad (v \neq 0), \qquad 0 \longmapsto 0.$$

Its radius increases strictly and continuously from 0 to π, so it is a bijection from the hemisphere to the closed angle ball. Continuity at $v = 0$ follows because the image norm is $2 \arcsin |v| \to 0$. By Proposition 2.9, composing it with the angle-ball rotation identifies only antipodal boundary points. Thus the induced map $\mathbb{R}\mathrm{P}^3 \to \mathrm{SO}(3)$ is a continuous bijection and is a homeomorphism by Corollary 2.4. The displayed rotation formula is therefore the antipodal covering followed by this homeomorphism, and hence is itself a covering $S^3 \to \mathrm{SO}(3)$. Its radial coordinates differ from Lecture 2's linear hemisphere rescaling.

The *special unitary group* $\mathrm{SU}(2)$ consists of complex unitary 2×2 matrices of determinant one. Its matrices have the form below, identifying it with S^3:

$$U(a, v) = \begin{pmatrix} a - \mathrm{i}v_3 & -v_2 - \mathrm{i}v_1 \\ v_2 - \mathrm{i}v_1 & a + \mathrm{i}v_3 \end{pmatrix}, \qquad a^2 + |v|^2 = 1.$$

Indeed, orthonormal columns and determinant one give the form $\left(\begin{smallmatrix} z & w \\ -\bar{w} & \bar{z} \end{smallmatrix}\right)$ with $|z|^2 + |w|^2 = 1$. Taking real and imaginary parts gives the displayed coordinates and a continuous inverse. This recalls the spinor cover of *Symmetries in Physics*, Lecture 14, in explicit coordinates.

The lifted path from the identity for the chosen rotation is

$$(a(t), v(t)) = (\cos \pi t,\ \sin \pi t\, e_3), \qquad U(t) = \begin{pmatrix} e^{-\pi \mathrm{i} t} & 0 \\ 0 & e^{\pi \mathrm{i} t} \end{pmatrix}.$$

At $t = 1$ it ends at $(-1, 0)$, or $-\mathbb{1}$ in matrix form. Its projection is closed, but its lift is not. By Corollary 4.4, this is the nontrivial loop class in $\mathrm{SO}(3)$.

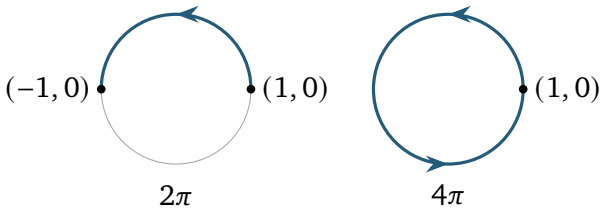


Figure 4.5: A full turn lifts from $(1, 0)$ to $(-1, 0)$ along a semicircle in S^3. A double turn completes the great circle and closes upstairs.

For a 4π rotation, replace t by $2t$ in the lifted formula. The lift ends at $(1, 0)$ and is a loop in the simply connected sphere. It contracts there, so its projection contracts in $\mathrm{SO}(3)$. In Dirac's belt experiment one end of a belt is fixed and the other is turned; a double full turn can be removed by moving the belt with its ends fixed. The covering calculation gives the corresponding loop-class statement for rotations. A spin-1/2 state in $\mathbb{C}^2$ transforms by $U(t)$: one full turn multiplies the state by -1, and two full turns multiply it by $+1$.

Self-check. Classify the connected coverings of the circle up to covering isomorphism.

Answer. The circle satisfies the classification theorem's hypotheses: small open arcs are intervals. Every subgroup of $\mathbb{Z}$ is $m\mathbb{Z}$ for exactly one integer $m \geq 0$. For a nonzero subgroup, its least positive element m and integer division prove this; the zero subgroup is $0\mathbb{Z}$. For $m \geq 1$, the map $z \mapsto z^m$ on S^1 induces multiplication by m on winding, so its image subgroup is $m\mathbb{Z}$. The cover $\mathbb{R} \to S^1$ has subgroup zero because $\mathbb{R}$ is contractible. The cited classification proves these exhaust the connected coverings. All subgroups are their own conjugacy classes, since $\mathbb{Z}$ is abelian.

Summary. Local inverse sheets lift paths and their homotopies uniquely. For a simply connected covering space, based loop classes correspond to fibre points. A transitive deck group gives the same group structure after choosing a lifted base point. The antipodal cover computes $\pi_1(\mathbb{R}\mathrm{P}^n) \cong \mathbb{Z}_2$ for $n \geq 2$, using the sketched sphere result. In $\mathrm{SO}(3)$, a full turn is nontrivial and a double turn is trivial. Lecture 5 computes fundamental groups by joining open pieces.

Reading. Hatcher, *Algebraic Topology*, Proposition 1.14, §1.3, especially Propositions 1.30–1.34 and Theorem 1.38; Munkres, *Topology*, §§53–54. Nakahara, *Geometry, Topology and Physics*, §4.6.5 and Table 4.1.